

Abdelrhman Hesham Galal

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SUMMARY

AI/ML and Backend Engineer specializing in Aviation Information Systems, combining machine learning and NLP with production backend engineering (Node.js, Express.js, MongoDB). Co-founder and AI Manager at Nowarha, leading AI initiatives for IoT solutions. Built a DistilBERT-based sentiment analysis system for airline passenger retention, achieving 93.5% accuracy and 89.18% Macro-F1, and single-handedly engineered a 40+ endpoint backend for the AI-powered airport platform Gate Buddy.

EDUCATION

Bachelor of Applied Science in Computer Science (Specialization: Aviation Information Systems)

Zagazig National University (ZNU) — Sept 2022 – June 2026

GPA 3.922/4.0 · Ranked 6th of 108 graduates

EXPERIENCE

Nowarha — Co-Founder & AI Manager | Egypt | 2026 – Present | [Nowarha Official Facebook Page](#)

AI-powered home electricity intelligence platform (IoT + AI). 8-person founding team.

- Co-founded Nowarha and lead AI development, driving model strategy and deployment for an AI-powered home electricity intelligence platform.
- Led AI/ML initiatives that earned 1st Place at Creativa Zagazig University and 3rd Place at the NextGen Hackathon (200+ projects).
- Built a three-model AI pipeline for load forecasting, bill prediction, and energy anomaly detection using household consumption data.

Digital Egypt Pioneers Initiative (DEPI) — Machine Learning Intern | Oct 2024 – Jun 2025 · 6 months

- Built the recommendation engine and CV extraction module for Jobotic, a full-stack AI career platform that placed 1st among ~300 projects at the CLS/DEPI graduation competition. Implemented TF-IDF and cosine similarity for candidate–job matching, a Random Forest classifier for fit scoring, and spaCy with regex for resume parsing.
- Completed a 6-month intensive training program covering machine learning, deep learning, and NLP, working with datasets exceeding 50,000 rows.

CodeAlpha — AI Intern | Remote | July 2026

- Built a real-time object detection and tracking pipeline (YOLOv8 + ByteTrack, OpenCV) processing live video with per-object ID persistence across 80 COCO classes.
- Built a Streamlit translation tool (100+ languages) with a custom slang-expansion preprocessor and gTTS text-to-speech.

Gate Buddy — Team Leader & Backend Lead | Zagazig University | Jan 2025 – Present

- Led an 8-person team building an AI-powered airport assistance platform; owned backend architecture, database, APIs, and AI integration.
- Coordinated backend, frontend, and mobile developers, defining API contracts and integration points to keep the team aligned from design through delivery.

Aviation Systems Internships | EGSA · NANS · EgyptAir | 2023

- Egyptian Space Agency (EGSA) | July 2023: Explored CubeSat development, satellite subsystems, and space technology workflows.
- EgyptAir | Aug 2023: Built a web/database system and gained exposure to airline operational processes.
- National Air Navigation Services Company (NANS) | Sept 2023: Studied radar systems, air navigation technologies, and CNS/ATM operations, including an on-site visit to the Cairo air traffic control (ATC) tower.

PROJECTS

Analyzing Airline Brand Loyalty Drivers: A Transformer-Based Causal Inference and Sentiment Framework | PyTorch · Hugging Face · NLP

- Fine-tuned DistilBERT models on 7,277 airline reviews to extract sentiment across four service dimensions (Cabin Crew, In-Flight Entertainment, Onboard Retail, and Overall Recommendation), achieving up to 93.5% accuracy and 89.18% Macro-F1.
- Applied causal inference methods (PSM, TWFE, and causal impact analysis) to quantify service drivers of airline brand loyalty and passenger recommendation behavior.

Gate Buddy — Node.js · Express.js · MongoDB · FastAPI · FCM | [Live Demo](#) · [Code](#)

- Built a 40+ endpoint REST API using Node.js, Express.js, and MongoDB covering authentication, flight tracking, airport services, and notifications.
- Implemented IATA BCBP boarding-pass parsing and automated flight tracking with live status synchronization, weather updates, and passenger alerts.
- Developed geospatial airport navigation services using MongoDB 2dsphere queries and Dijkstra-based routing for terminal wayfinding.
- Integrated a FastAPI recommendation microservice for destination discovery with automated scraping, ranking, and caching.

Aviation Safety Hub — React · Electron · Gemini | [Code](#)

- Developed an aviation decision-support system that digitizes pre-flight risk assessment using IMSAFE, PAVE, and DECIDE frameworks to help pilots identify hazards and make safer Go/No-Go decisions.
- Implemented a cumulative risk scoring engine and GRM risk matrix to evaluate pilot, aircraft, and environmental factors with structured mitigation recommendations.
- Integrated an AI Pilot Advisor that analyzes assessment results and generates personalized safety insights while maintaining secure API handling.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, SQL

ML/AI: PyTorch, HuggingFace/Transformers, Scikit-learn, NLP, Deep Learning, Model Evaluation

Backend: Node.js, Express.js, MongoDB, Mongoose, REST APIs, JWT/OAuth

Data: Pandas, NumPy, Matplotlib, Seaborn, Plotly

Tools: Git, Postman

CERTIFICATIONS

- Udemy – Complete Node.js Developer Course (Jonas Schmedtmann, 2025): backend development with Node.js, Express, MongoDB, Mongoose; built the Natours full-stack app.
- DEPI Machine Learning Internship Certification (2024).
- Kaggle Pandas Certification (2024).
- HackerRank Python (Basic) Certification (2023).

ACHIEVEMENTS

- 1st Place — CLS/DEPI Graduation Competition (Jobotic AI career platform; 1st of ~300 projects)
- 1st Place — Creativa, Zagazig University (Nowarha; 5-judge panel)
- 1st Place — Sumo Robotics Competition (32-team single-elimination bracket)
- 3rd Place — NextGen Hackathon (Nowarha; 200+ competing projects, 3-judge panel)
- Top 25% Globally — Energy Mentors Competition (231 teams)